

Ramana Gangarao

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Experience

AI Engineer

San Francisco, CA

Optum

Dec 2025 – present

- Implemented an Analytics Agent using Vertex AI (Gemini) and Google ADK, implementing the Model Context Protocol (MCP) to standardize tool-to-model communication and enable plug-and-play integration with the Looker API.
- Engineered a Multi-Agent Orchestration Framework with a Hierarchical Memory System, utilizing Redis for session-based short-term context and a Vector Database for long-term retention of user preferences and recurring KPI definitions.
- Implemented a Chain-of-Responsibility Fallback Architecture, where a primary reasoning agent triggers a Human-in-the-Loop intervention if confidence scores validated by the LLM-as-a-Judge fall below a defined threshold.
- Developed a State-Aware Metadata Discovery Agent that crawls Looker metadata (Models, Explores) integrated Short-term Memory to prevent redundant API calls and reduce token consumption.
- Engineered Type-Safe Agent Tools via MCP and Pydantic, enforcing strict JSON schema validation and creating a deterministic interface for the agent to execute complex data-shaping functions without parsing failures.
- Deployed Event-Driven Agentic Microservices on GCP (GKE, Pub/Sub) that support Asynchronous Multi-Step Reasoning, enabling agents to maintain state across long-running data processing tasks and automated anomaly detection.
- Established Deep Agent Observability using OpenTelemetry and Cloud Logging to trace agent reasoning loops, monitor Memory Retrieval Latency, and audit fallback triggers to continuously refine agent prompts and tool accuracy.

AI Engineer

Dublin, CA

Innovapath INC - TalentScreen

Jan 2025 – Dec 2025

- Developed AI powered talent management platform using RAG and multi-agent orchestration to standardize hiring, reduce bias, and cut time-to-hire.
- Designed a secure, multi-tenant AI architecture with strict tenant isolation across vector stores, databases, memory layers, and agent execution contexts.
- Implemented hierarchical multi-agent orchestration using Google ADK, enabling deterministic agent routing, multi-tool execution, and extensible agent lifecycles.
- Developed intent classification and routing logic to dynamically dispatch user queries to specialized agents (Recruiter, Hiring Manager, Mobility, Policy).
- Applied the planner-executor agentic pattern to decompose complex tasks into structured plans and executions, improving output accuracy and reliability.
- Designed and optimized the RAG pipeline with document chunking strategies, metadata-aware retrieval, hybrid semantic filtering, and reranking to improve retrieval precision.
- Stored end-to-end agent execution and conversational data in GCP NoSQL persistence (Firestore) to meet audit, replay, and compliance explainability requirements for hiring decisions.
- Integrated Langfuse for end-to-end tracing, capturing prompt versions, tool calls, retrieval context, agent transitions, and latency metrics.
- Conducted system-level and agent-level evaluations using task benchmarks, LLM-as-judge scoring, and ablation studies (agent removal and fault injection) to assess robustness, emergent behaviors, error propagation, and performance gains over single-agent baselines.
- Implemented hybrid memory management with short-term session memory (Redis) and long-term persistent memory using PostgreSQL and Milvus for RAG context.
- Built comprehensive guardrails including grounded-response enforcement, citation validation, bias masking, schema-validated outputs, and hallucination detection.
- Architected and deployed agentic AI services on GKE, enabling scalable, containerized orchestration of multi-agent workflows.

- Used Vertex AI for managed LLM inference, evaluation workflows, and model versioning across development and production environments.
- Implemented centralized observability and security using Cloud Logging, IAM, and Secret Manager for auditability, access control, and secrets management.

Developer Analyst I / MLOps

Wipro Limited

Chennai, India

Sept 2021 – Aug 2024

- Designed and maintained offline and online feature pipelines, including schema validation, time-based aggregations, and consistency between training and serving data.
- Implemented real-time feature updates using incremental writes and TTLs to keep features fresh and production-aligned.
- Deployed and operated real-time model serving APIs on Amazon SageMaker, managing authentication, versioned endpoints, traffic routing, and rollback strategies.
- Built scalable batch inference pipelines for large datasets with idempotent outputs and full input/output traceability.
- Integrated A/B testing and shadow deployments to safely evaluate new models against live production traffic.
- Used MLflow for experiment tracking, logging parameters and metrics, and managing model versions via the model registry.
- Automated model packaging and deployment through CI/CD pipelines, promoting models across dev, staging, and production environments.
- Implemented monitoring and alerting for model latency, error rates, and data drift to detect production issues early.
- Supported automated retraining workflows, triggering pipelines based on performance degradation or data quality signals.
- Collaborated with data science and platform teams to define model contracts, SLAs, and production readiness standards.

Software Engineer

INNOVAPATH IT Solutions PVT Ltd

Hyderabad, India

Sept 2016 – Sept 2021

- Built and deployed Spring Boot microservices using JPA/Hibernate, applying design patterns and Java 8 features for scalable backend development.
- Developed automation frameworks using Selenium, TestNG, RESTAssured, and BDD, improving UI and API test coverage and maintainability.
- Containerized Java and Python services with Docker and deployed on AWS EC2 and S3, ensuring high availability and production stability.
- Implemented CI/CD pipelines using Jenkins and Git, automating builds, tests, and deployments for faster and reliable software delivery.

Skills

Programming Languages: Python, JavaScript, Java, SQL, LookML

AI & Agentic Libraries: Google ADK, LangChain, LangGraph, LlamaIndex, Pydantic, Langfuse, HuggingFace, Transformers

LLMs & AI Platforms: Vertex AI (Gemini), OpenAI, Anthropic Claude, LLaMA, Mistral, AWS Bedrock

Machine Learning & Frameworks: PyTorch, Scikit-learn, MLflow, NumPy, Pandas, React.js, Node.js, Spring Boot

Databases & Vector Stores: PostgreSQL, MongoDB, MilvusDB, Neo4j, RDS

Cloud & DevOps (GCP & AWS): GCP (Cloud Run, GKE, Pub/Sub, Secret Manager), AWS (EKS, Lambda, S3, EC2), Docker, GitHub Actions

Education

Atlantis University

MS in Information Technology

Miami, FL, USA

Aug 2024 – Oct 2025

Keshav Memorial Institute of Technology - JNTUH

Bachelor of Technology in Electronics and Instrumentation Engineering

Hyderabad, India

July 2013 – Apr 2016